

Total No. of Questions : 8]

SEAT No. :

[Total No. of Pages : 2

PA-1611

[5926]-245

T.E. (AI & DS)

PATTERN RECOGNITION

(2019 Pattern) (Semester-I) (Elective-I) (317522B)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn whenever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) Clarify the Chomsky Normal form with suitable example? [9]
b) Describe the Grammar Based Approach and its Application in detail.[8]

OR

- Q2)** a) Describe any two examples of String Generation as Pattern Description?[9]
b) Give an Abstract view of parsing problem? [8]

- Q3)** a) Compare Homomorphism and Isomorphism? [9]
b) Analyze different application of Relational Graph to Pattern Recognition?[8]

OR

- Q4)** a) Describe Clique finding algorithm with suitable example? [8]
b) Describe Canonical definite finite state grammar (CDFSG) with suitable example? [9]

- Q5)** a) Explain CAM & other Neural Memory Structure? [9]
b) Describe Neural Networks as a Black Box Approach? [9]

OR

P.T.O.

- Q6)** a) Describe with neat diagram Artificial Neuron Activation and output characteristics? [9]
- b) What are the different reasons to adopt a Neural Computational Architecture? [9]

- Q7)** a) Explain the structure of a Multiple Layer Feedforward Network? [9]
- b) Draw & explain how to train the feedforward network using Generalized delta Rule? [9]

OR

- Q8)** a) Describe how the character classification is done with pattern Associator?[9]
- b) Draw & Explain Summary of the Back Propagation learning Procedure?[9]

